ICT2152 · Object Oriented Analysis and Design

Core Concepts: UML Class and Sequence Models

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

A class diagram shows types, attributes, operations, and relationships. A sequence diagram shows collaborating objects and messages over time. Multiplicity communicates how many objects may participate in a relationship.

2. Example and reasoning

An offering may have many enrollments and many materials. A sequence diagram can show a screen sending a request to a controller, the service checking authorization, and a DAO retrieving scoped data.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “core concepts: uml class and sequence models” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.